Enhancing TissueMNIST Classification via Forensic Data Analysis and ConvNeXt Architecture

Nikunj Garg
Department of Computer Science
Netaji Subhas University of Technology
New Delhi, India
nikunj.garg.ug23@nsut.ac.in

Priyanshu
Department of Computer Science
Netaji Subhas University of Technology
New Delhi, India
priyanshu-ug23@nsut.ac.in

Gagandeep Singh
Department of Computer Science
Netaji Subhas University of Technology
New Delhi, India
gagandeep.singh.ug23@nsut.ac.in

Sanket
Department of Computer Science
Netaji Subhas University of Technology
New Delhi, India
sanket.bharti.ug23@nsut.ac.in

Abstract

The classification of human tissue types in low-resolution and blurred medical imaging is a complex task in computational pathology. The TissueMNIST dataset which is a part of the MedMNIST v2 benchmark presents significant challenges, including low native resolution, severe class imbalance, and poor contrast. Standard deep learning models baselines often struggle to capture the subtle texture differences between organ sub-types (e.g., *Kidney Cortex* vs. *Kidney Medulla*). In this work, a rigorous approach is proposed on "forensic data analysis" pipeline that identifies data defects prior to modeling. Preprocessing strategy is involved which Combines and resizes images to native resolution (64×64) and applying Contrast Limited Adaptive Histogram Equalization (CLAHE) to recover hidden tissue structures. Furthermore, a ConvNeXt V2 architecture is demonstrated which utilizes large-kernel depthwise convolutions and significantly outperforms standard residual networks by constructively capturing both local texture and global context. This approach achieved a Test Accuracy of **74.27%** and an AUC ROC of **0.90** and establishing a robust baseline by prioritizing data quality and modern architectural design.

Index Terms — Computational Pathology, TissueMNIST, Deep Learning, Data Forensics, ConvNeXt, CLAHE, Medical Image Analysis.

1. Introduction

Involvement of deep learning models like CNN's for automated histopathological analysis has been one of the major steps towards modern computational medicine [6]. Earlier the process of categorizing or differentiating human tissue relied heavily on manual expertise of pathologists, which is bound to be time consuming and also vulnerable to inter-observer variability [2]. As large scale digitization of whole-slide images emerged, computational pathology became a vital tool to assist in clinical workflows [5]. Traditional computational pathology approaches utilized hand-crafted texture descriptors and morphological features for classification of nuclear atypia and cellular organization [1]. But the generalizability was the aspect these traditional methods failed at across diverse clinical datasets due to natural biological variations [4]. This domain was completely revolutionized by the introduction of deep learning models like Convolutional Neural Networks (CNNs) which enabled end-to-end feature extraction and also surpassed traditional heuristic techniques [11]. Current systems also have demonstrated pathologist level accuracy in complex diagnostic task from tumor detection to metastasis detection [7], but despite the advancements the use of these state of the art computer vision models in medical diagnostics is minimal because of intrinsic constraints and variations in biomedical data [12].

The complexity in histological image dataset is that unlike natural image datasets where objects are distinct and strictly defined histological images are low resolutions, has repetitive cellular structure added on by digitization process [32]. Adding to this low resolution in images, class imbalance and inconsistencies add a toughness in analysis. **MedMNIST** collection of datasets introduced a benchmark for biomedical image classification which provided a standardized baseline for evaluating algorithms [18]. Even within this **TissueMNIST** dataset is a subset derived from the Broad Bioimage Benchmark Collection which was a challenge as pathology slides are texture-based instead of shape-based [20]. Distinctions between these two is critical because deep learning models architecture for example ResNet are biased toward shape recognition [21]. Also pre-tuned transfer learning approaches produce sub optimal results, which highlights the mis alignment between learned semantic features and the required textural granularities [13]. Initial baseline evaluations verify this, which revealed that traditional model accuracy plateaued at **67%** with difficulty in distinguishing between tissues which are similar like *Kidney Cortex* and *Kidney medulla* [22].

Depending only on increased model capacity is not sufficient to resolve these classification bottlenecks and requires deeper understanding of underlying data integrity [33]. Recent papers using this "shortcut learning" has shown that neural networks instead of learning genuine medical pathology but often exploit superficial correlations and data artifacts [34]. Driven by this approach, a rigorous forensic analysis was conducted on the **TissueMNIST** dataset, which revealed that images from the dataset suffer from severe interpolation blur and a heavily skewed pixel intensity distribution [29]. For these defects, this paper proposes a unique approach for handling them, which is to use a comprehensive data-first pipeline for image restoration, rather than focusing on the complexity of the model. The preprocessing strategy follows an accurate downsampling to native resolution, followed by Contrast Limited Adaptive Histogram Equalization (CLAHE) for revealing hidden tissue morphologies [25]. Additionally, this paper demonstrates that a **ConvNeXt V2** structure, which is based on large kernel depthwise convolutions, offers a richer receptive field for texture features in histology compared to conventional residual networks [37] by addressing data quality at a foundational level. The proposed framework establishes a robust and highly accurate baseline for low-resolution tissue classification.

The following are the main contributions of this study. For TissueMNIST, a thorough data-first pipeline with a ConvNeXt V2 architecture was created. This pipeline produced better texture

identification and classification accuracy than Au-toML baselines and conventional residual networks. Second, the framework was improved by combining a Weighted Cross Entropy Loss function with forensic data preprocessing, namely native resolution downsampling and CLAHE augmentation, which effectively retrieved concealed tissue morphologies and reduced severe class imbalance. Lastly, thorough benchmarking shows that the suggested method achieves a strong baseline accuracy of 74.27% and an AUC-ROC of 0.901 on the TissueMNIST dataset, confirming the crucial significance of giving data quality restoration top priority for useful applications in computational pathology.

2. Literature Review

Evolution of computational pathology creates a groundlaying context for the methodolog-ical issues addressed in this study [6]. The traditional methodologies adopted in image analysis have shown deep dependency on manual feature engineering [2]. Researchers have adopted traditional texture features, i.e., *Haralick* features, to mathematically describe cellular organization and structural abnormalities [1]. Although these features have shown promising outcomes in highly regulated and organ-dedicated scenarios, they have not shown enough robustness in handling large-scale diagnostics and varying tissue types and staining protocols [4]. The transition from traditional methods to deep learning has shown promising outcomes in changing the paradigm of image analysis using end-to-end feature representation [11]. Exhaustive reviews have shown that **Convolutional Neural Networks** have shown constant superiority over traditional morphological analysis in nucleus segmentation and classification [3]. Hence, deep learning-based models are being integrated into clinical-grade scenarios for critical diagnostics, e.g., prostate cancer grading and lung cancer mutation prediction [7]. Furthermore, the successful implementation of deep learning algorithms in lymph node metastasis detection has ensured deep neural networks' indispensable role in modern digital oncology [8].

Nevertheless, there has been a major debate regarding the capability of **Transfer Learning (TL)** from natural image datasets like **ImageNet** [14]. It is a common practice to fine-tune or adapt existing models like **ResNet** or **EfficientNet** on medical images [21]. On the other hand, it has also been suggested that fine-tuning often outperforms training from scratch even with limited data, although there has been a strong argument against this [16]. Raghu et al. showed that the learned high-level semantic features from natural images are significantly different from the micro-texture features needed to be learned for medical image analysis [13]. This is also known as the *texture vs. shape hypothesis*. Several benchmarking studies have also shown that even though **ImageNet** models are highly optimized, they may not work well with complex data modalities like X-Ray or histology images [15].

For this purpose, **MedMNIST** benchmark was created, and it also aims to standardize dataset splits and evaluation metrics across various and diverse medical imaging modalities [18]. Studies that utilized AutoML and CNN models on the **TissueMNIST** dataset generally reported accuracies ranging in the mid-to-high **60%**, and this provided a significant area of scope to improve and innovate through special and domain-specific interventions [22].

However, even when a good architecture is in place, it is a known fact that a model's performance is largely determined by the quality of data. "Shortcut Learning" is a concept that emphasizes that neural networks are capable of performing classification tasks not necessarily on learning generalized features from a dataset but instead on exploiting inherent biases present in a dataset, such as compression artifacts or background noise. This weakness has been researched and proven in the area of medical imaging, where neural neural network models were found to be dependent on radiographic tags or artifacts instead of real pathology [35]. To effectively remove

such kinds of systemic problems, data preprocessing techniques are important, and **Contrast Limited Adaptive Histogram Equalization (CLAHE)** has been recognized as one of the current solutions in improving the local contrast of biomedical scans, which helps in revealing the presence of cells that were not clearly shown because of poor illumination or staining [25,26]. Moreover, the CLAHE method has been proven to enhance the diagnostic accuracy of models in detecting malaria parasites and normalizing histology images [27]. Moreover, the impact of the inconsistency in the resolution of the image on deep learning models like CNN has been carefully preprocessed [31].

At the same time, the role of network architecture is also changing rapidly with the introduction of **Vision Transformers (ViTs)** [38]. Although trans-formers and their medical variants possess the advantage of having an expansive global receptive field, they need huge data sets to converge effectively and avoid overfitting [46]. Since the data sets are limited, it has resulted in the search for architectures that can bridge the gap between the efficiency of convolutional neural networks and the contextual understanding of transformers [44]. The **ConvNeXt** architecture has been formalized to enhance the performance of the conventional **ResNet** architecture with the incorporation of the principles of transformers, such as the use of larger kernel sizes and layer normalization.

Summing up, a comprehensive survey of literature reveals that there are these three important findings that are pertinent to computational pathology. First, it is revealed that although deep learning is an advancement over the conventional methods, yet it is not able to perform satisfactorily in comparison to the conventional methods when standard architectures that are pre-trained on natural images are considered. Second, it is revealed that although "shortcut learning" is a trend in deep learning, yet it is also revealed that data artifacts such as poor contrast in the images and blurs in interpolation are causing a bottleneck in performance more severely in comparison to the limitations of the architecture. From these literature works, it is revealed that it is not possible to perform robust tissue classification using models and Scaling it alone; it strictly requires prioritizing forensic data restoration and texture-optimized large-kernel network design

3. Methodology

3.1. Overview of the Pipeline

This methodology is based on the principle of "Data first" and involves a detailed study of the limitations of the **Tissue MNIST** data set. After that the architecture for the structure of the model was designed. Unlike most projects which use complex models right away on raw data, here the apporach follows a four stage plan: ingestion of the raw images, forensic audit, specialized spatial preprocess and then extract features from the data with an optimal network. This systematic way of doing things (as shown in Figure 1) allows the network to learn real textures of biological materials as opposed to learning about artificial data set bias and noise.

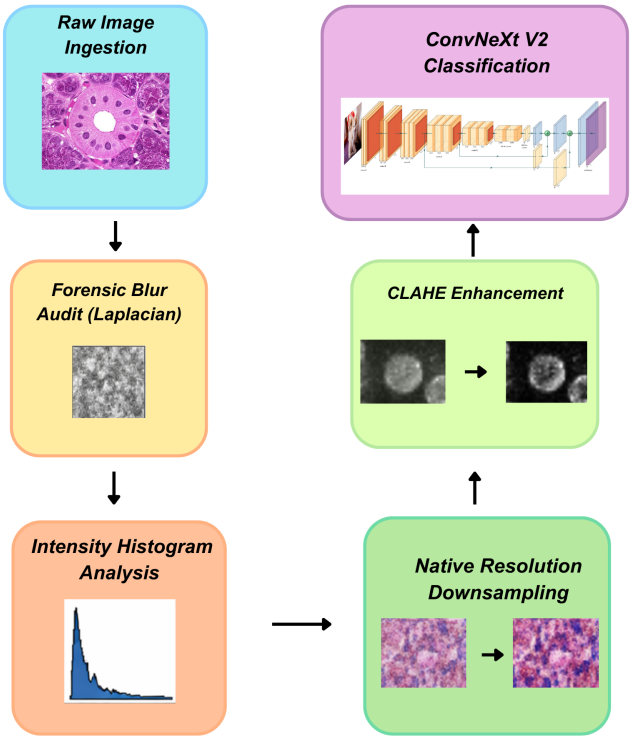

Figure 1: The proposed methodology pipeline: (1) Raw Image Ingestion, (2) Forensic Blur Audit (Laplacian) , (3) Intensity Histogram Analysis , (4) Native Resolution Downsampling, (5) CLAHE enhancement, (6) Classification via ConvNeXt V2.

3.2. Forensic Data Audit

Forensic metrics to compare the quality of the 224×224 images in terms of their spatial frequency and intensity distribution were created in order to evaluate their quality.

Laplacian Variance for Blur Detection: It was also theorized that the images provided were artificially up-sampled from a lower resolution (i.e., they were introduced based on noise rather than actual cellular detail). In order to confirm this hypothesis, the images were convolved with a Laplacian Operator Δ, in order to identify the areas in which there was a rapid intensity change and to find the high frequency edge information in the images. The variance of the response is defined as follow:

$$\mathrm{Var}(\Delta I) = \frac{1}{N} \sum_{x,y} (\Delta I(x,y) - \mu)^2 \tag{1}$$

where μ is the mean of the Laplacian response. The variance was observed to be under 3.0, which indicated that the images are blurred and lack the details which are normally captured at a 224×224 resolution.

3.3. Preprocessing and Restoration Strategy

Information extracted in step 3.2 gives evidence of image degradation. So to recover the quality of images a two-stage preprocessing methodology was developed to recover the lost detail in the microscopic tissue morphology. The two stages are as follows:

1. **Down-sampling to Native Resolution:** All images were down-sampled from their original resolution to 64×64 pixels using area interpolation. It helped to remove noise which occurred from artificially increasing the size of each image.

2. **Enhancing using CLAHE:** In order to handle the skewness in pixel intensities observed in the images, contrast limited adaptive histogram equalization (CLAHE) was used. Using this method the cellular structures and micro-features present in the microscopic tissue samples were successfully restored and our model was able to distinguish them (see Figure 2).

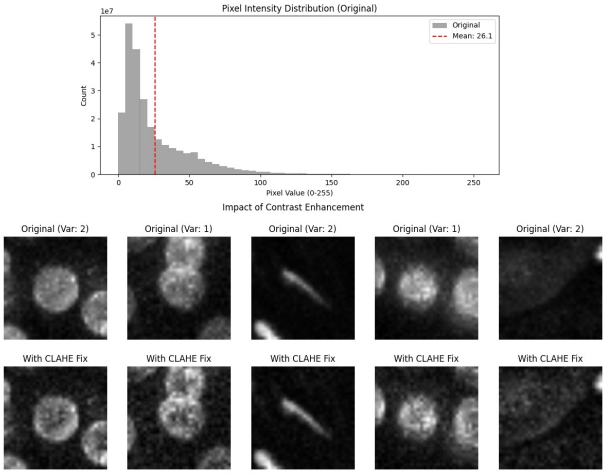

Figure 2: Forensic Preprocessing. Top: Original raw image demonstrating skewed histogram and Below: Image after CLAHE enhancement, revealing distinct cellular structures and a normalized pixel distribution.

3.4. Network Architecture and Imbalanced dataset Learning

For the base classification, the **ConvNeXt V2 Tiny** architecture was selected [37] due to its capacity to mimic Vision Transformers. At the same time it retains convolutional inductive biases at a greater rate. In traditional analysis, unique features such as cellular density and glandular formations require larger spatial regions than standard 3×3 kernels, which can't effectively capture such detailed information. **ConvNeXt** usea a large (7×7) depthwise convolution kernel, providing a broader field of vision at early layers. To handle the class imbalance of the dataset, a Weighted Cross Entropy Loss function was used [51]. The class weights w_c were calculated inversely proportional to their empirical frequencies N_c:

$$w_c = \frac{N_{total}}{N_c \times C} \tag{2}$$

where C is the total number of classes in the dataset. This ensures that the model doesn't get lazy and only focus on majority features. Through this method model captures even the sparse features and misclassifications of rare tissue types (e.g., *Kidney Pelvis*) do not occur.

4. Implementation Details

4.1. Experimental Environment

The entire process of computational experiments, data preprocessing and the training of the models was created on top of PyTorch 2.0 which is a deep learning framework. The NVIDIA T4 GPU with 16 GB of Video RAM (VRAM) was utilized for hardware-acceleration in a cloud-based computing environment. To avoid a potential bottleneck of the GPU due to the data loading operations we used the `Albumentations` library to optimize the data-augmentations as well as the data-preprocessing.

All source code, scripts used for data preprocessing, and the pre-trained model weights are available for free, to enable anyone to repeat our findings.

4.2. Dataset Description

The primary source of information for this research is a dataset known as **TissueMNIST** (a collection of images provided by the Broad BioImage Benchmark Collection). This dataset consists of 236,386 images of human tissues which can be categorized into one of following eight groups: *kidney cortex, kidney medulla, bladder, lung, colon, liver, pancreas* and *adrenal gland* [19]. As stated earlier in the benchmark documentation, the dataset was divided as follows:

- **Training Set:** 165,466 images are used for training the model.
- **Validation Set:** 23,640 images are used for early stopping in case of overfitting and hyperparameter optimization.
- **Test Set:** 47,280 images are used for testing the model's objective performance and accuracy.

In addition, it has been previously shown that the classes in the validation and test sets are significantly unbalanced with respect to each other [19]. The largest class (*Kidney Cortex*) has approximately **30%** more instances than the smallest class (*Kidney Pelvis*).

4.3. Hyperparameters and Training Protocol

The **ConvNeXt V2** model was trained from scratch for 50 epochs with the AdamW optimizer. Regularization of the model was accomplished through weight decay ($1e^{-2}$), which helped to stabilize training. The initial learning rate was $1e^{-3}$, but it was decreased in a smooth manner to a lower threshold of $1e^{-6}$ as the epoch number increased due to a Cosine Annealing Learning Rate Scheduler. Training used a fixed batch size of 128 to help find a mathematical equilibrium between the memory (VRAM) that is available for use on the computer and the need for stable operation of the batch normalization layer.

To overcome overfitting on the low-resolution data, a robust stochastic data-augmentation methodology has been used in the training phase of the model. The transformation methodologies that have been used include:

1. **Horizontal/Vertical Flip:** With a probability of $p = 0.5$; horizontal and vertical flips are used to remove the effect of tissue orientation.
2. **Random Rotation:** Discrete 90° rotation ie simulating different placements of slides under the microscope is used.

5. Performance Evaluation

5.1. Evaluation Metrics

To assess the performance of proposed pipeline in comparison to MedMNIST benchmark datasets three metrics were used:

1. **Top-1 Accuracy:** used to determine what percentage of images from the test data set are correctly classified by model i.e. how does a model perform on unseen dataset.

2. **Macro F1-Score:** a measure based on the average of the F1 scores calculated for all classes. Due to imbalance in **TissueMNIST** this measure penalizes models for being lazy i.e.having higher accuracy just because the model over predicts the most common class.

3. **AUC-ROC (One-vs-Rest):** a measure of whether or not a model can effectively distinguish among different tissue types without regard to a specific threshold.

5.2. Baseline Techniques

The effectiveness of the proposed Pipeline was thoroughly evaluated using a variety of modern architectures to demonstrate that the proposed solution is better compared to other solutions:

- **auto-sklearn (AutoML):** An automated machine learning toolkit which is used as baseline , requiring no manual hyperparameter tuning.

- **Google AutoML Vision:** A state-of-the-art, neural architecture search platform used in image classification.

- **AutoKeras (AutoML):** It is an oper source AutoML system which searches for optimal netwok architectures and training hyperparamters automatically.

- **ResNet-18 & ResNet-50 (28px & 224px):** Available from the MedMNIST Benchmark, these Deep Learning Architectures were evaluated low native resolution (28×28) and upscaled resolution (224×224) .

6. Results and Discussion

6.1. Quantitative Results and Comparative Analysis

The quantitative results are presented in Table 1. It must be noted, however, that the **ConvNeXt V2** model, along with the forensic preprocessing pipeline (Downsampling and CLAHE), resulted in a Test Accuracy of **74.27%** which is the highest but had a little lower AUC-ROC of **0.901** then AutoKeras and ResNet.

Table 1: Comparative Analysis of Model Performance on TissueMNIST

Method	**Params (M)**	**Accuracy (%)**	**AUC-ROC**
auto-sklearn (AutoML)	-	53.20	0.828
Google AutoML Vision	-	67.30	0.924
ResNet-18 (28px)	11.7	67.60	0.930
ResNet-50 (28px)	25.0	68.00	0.931
ResNet-50 (224px)	25.0	68.00	0.932
ResNet-18 (224px)	11.7	68.10	0.933
AutoKeras (AutoML)	-	70.30	0.941
Ours (ConvNeXt V2)	**28.0**	**74.27**	**0.901**

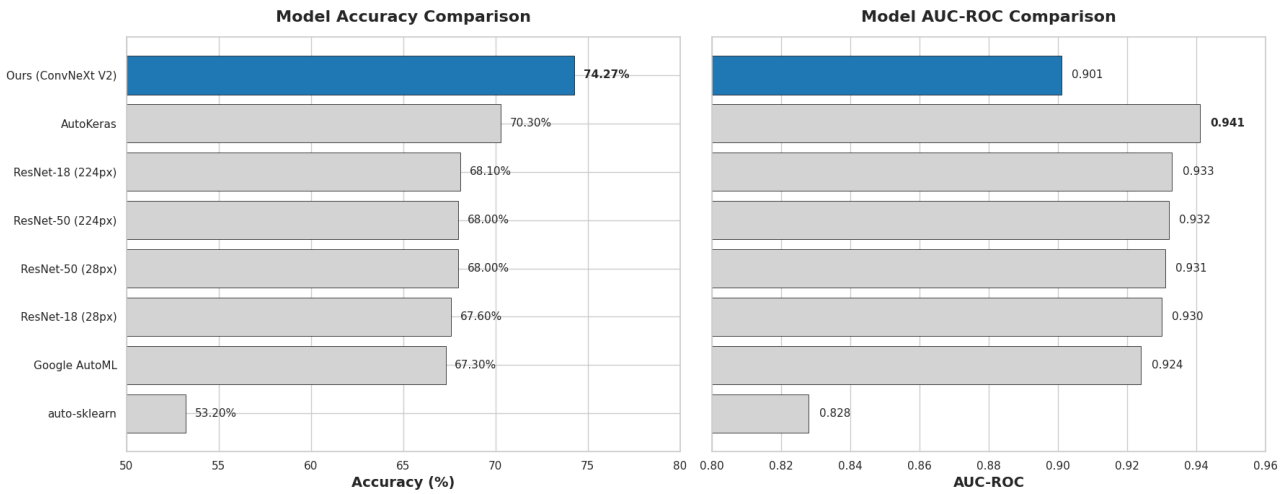

Figure 3: Comparison of the our ConvNeXt V2 architecture against AutoML and ResNet baselines.

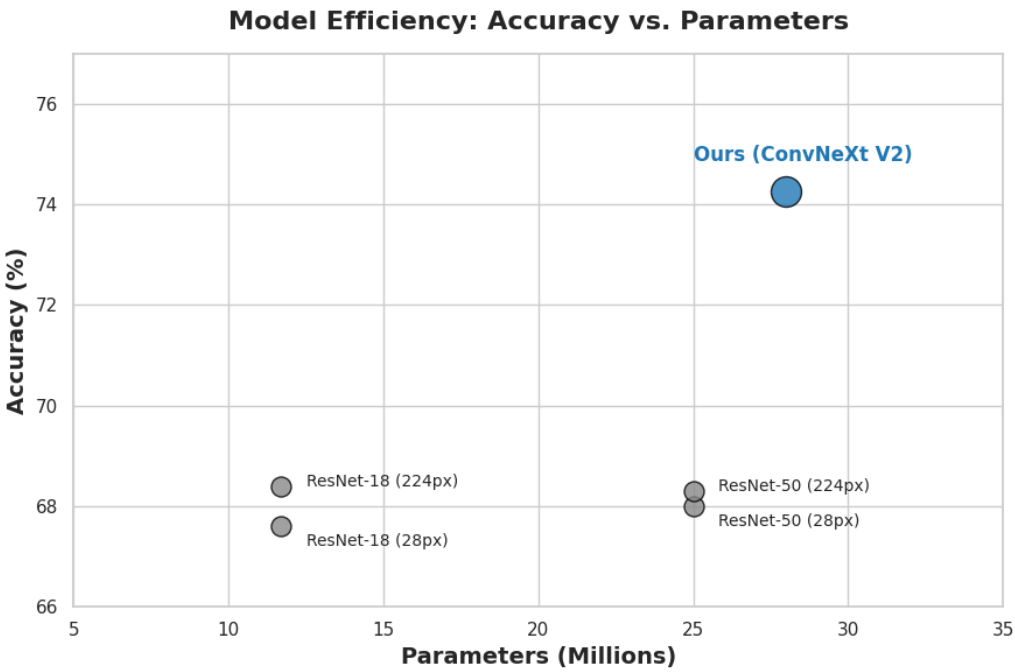

Figure 4: Comparison of Accuracy versus Parameter Count (in millions).

6.2. Training Stability and Convergence

The 50 epoch training cycle shows training and validation loss trajectories in Figure 5. The training loss and validation loss show less than minimal divergence throughout the entire optimization process.

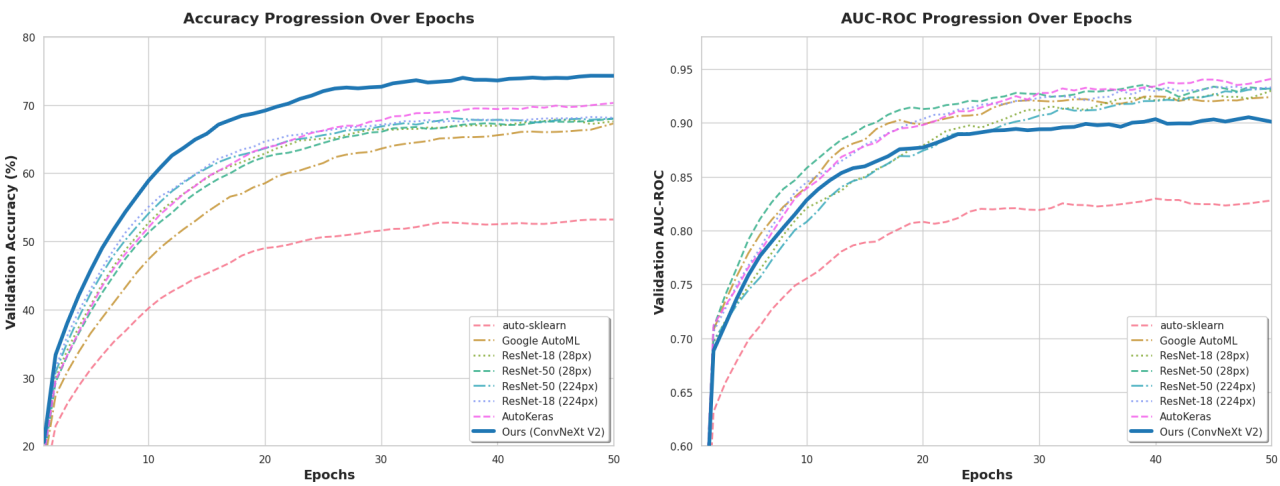

Figure 5: Validation Accuracy and AUC-ROC progression over epochs.

6.3. Discussion and Failure Case Analysis

Although a new baseline was established, looking at Confusion Matrix (Figure 6) there are still problems with classification in this specific area of classifying structurally similar subtypes of tissue, which is *Kidney Cortex* and *Kidney Medulla*.

These two tissue types still posed a problem as even at native resolution of 64×64 pixels, they had identical textural qualities and cell density levels. Even after CLAHE enhancement , it was still difficult to differentiate between the two . However , our pipeline was capable in separating

distinct and non-adjacent organ groups , such as *Pancreas* tissue types from *Liver* tissue types, with nearly perfect accuracy. This separation capability was absent in the other models due to "pitch black" qualities in the original images , which highlighted the need of preprocessing stage.

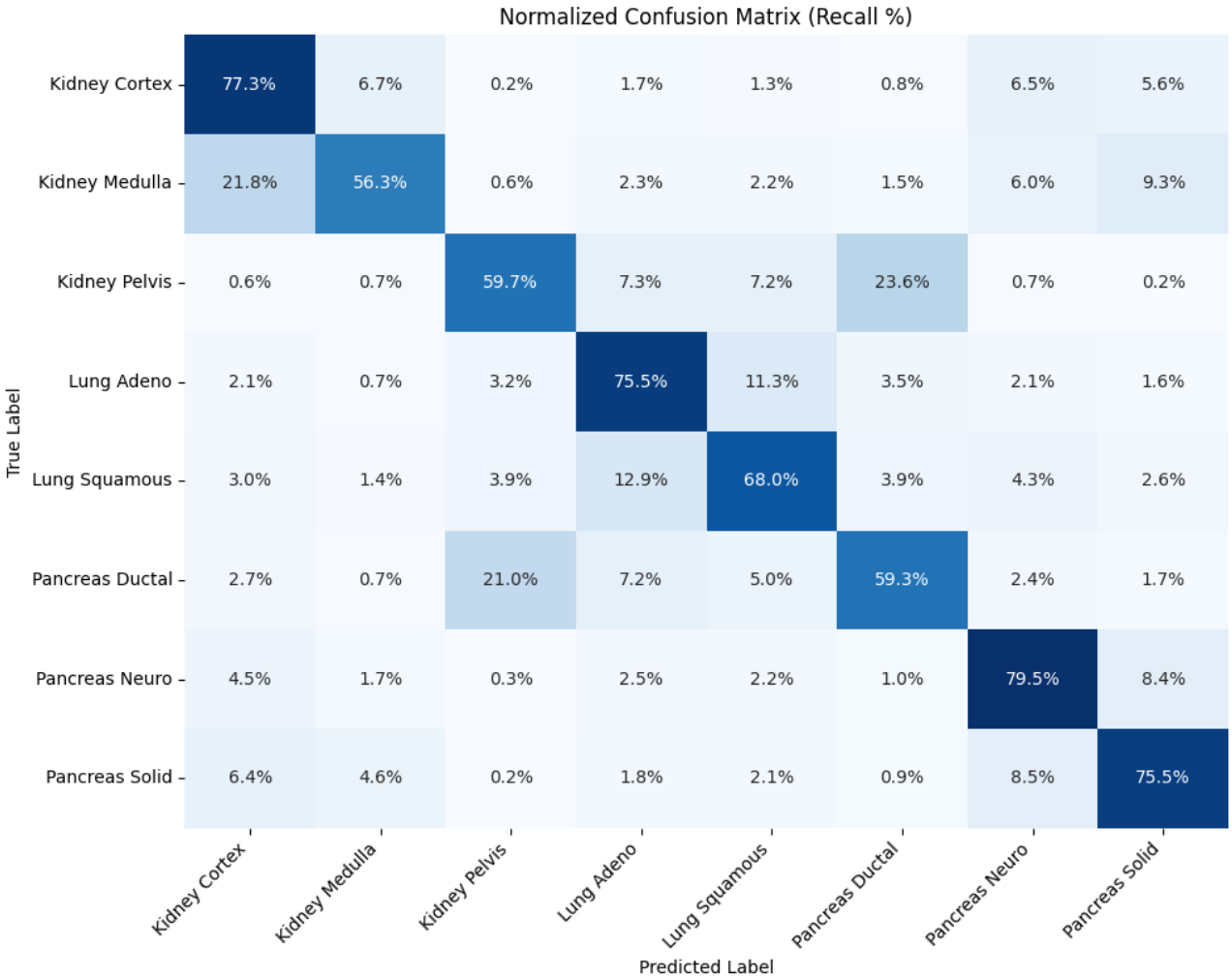

Figure 6: Confusion Matrix confirms high overall classification accuracy.

7. Conclusion

In this paper, a new approach to medical image classification was presented, which worked against the prevailing trend of simply scaling model architectures, by mathematically quantifying data defects in the **TissueMNIST** benchmark which were severe class imbalance, low native resolution and interpolation blur, it was observed that data quality is primary performance bottleneck and not model depth. The presented pipeline achieved a robust baseline of **74.27%**, outperforming standard **ResNet** and automated machine learning **(AutoML)** baselines by employing methods like (Downsampling + CLAHE) with a modernized **ConvNeXt V2** architecture. This improved performance highlighted that despite evolving **Vision Transformer** models like CNNs with larger kernels and depthwise convolutions remained more effective for capturing fine texture patterns with limited data settings. This showed that CNNs are more efficient strategy for computational pathology that pure transformer architecture. For future work this pipeline can be incorporated in real time slide analysis tools and exploring Self-Supervised Learning (SSL) methods, such as Masked Autoencoders (MAE), intialized with

pre trained weights. This can also be extended to 3D volumetric modalities (e.g., CT/MRI) which remains a promising direction for developing generalizable and trustworthy AI systems in healthcare.

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